

2. Parameters

Every arbitrary constant in this procedure is a named parameter listed here. The suggested values are starting points for the VSX3000 at 500–1000 mm. Values marked † depend on VSX3000 datasheet or SDK values I do not have, so they must be confirmed or replaced in Step 4.5.

Name	Suggested value	Units	Used in	Meaning
Sensor geometry				
SENSOR_FX_PX, SENSOR_FY_PX	from SDK †	px	all	Left IR focal lengths
SENSOR_CX_PX, SENSOR_CY_PX	from SDK †	px	all	Left IR principal point
SENSOR_BASELINE_MM	from datasheet †	mm	C, E	Stereo baseline B
PROJECTOR_OFFSET_MM	from datasheet †	mm (H, V, Z)	C, E	Projector center relative to the left camera
DEPTH_LSB_MM	from SDK †	mm	A	Least significant bit of the depth output
Working volume				
Z_MIN_MM / Z_MAX_MM	500 / 1000	mm	all	Range limits
Z_NOISE_STEP_MM	50	mm	A	Spacing of noise stations, giving 11 stations
Z_SHAPE_STATIONS_MM	500, 625, 750, 875, 1000	mm	B, C, D	Stations for edge, area, and detection tests
Z_REDUCED_STATIONS_MM	500, 750, 1000	mm	B-Z, D (0% series)	Subset used for the slowest tests
FIELD_OFFSET_FRACTION	0.6	of half-FOV	A, C	Off-axis field positions at ($\pm f \cdot W/2$, $\pm f \cdot H/2$) from center
TILT_ANGLES_DEG	0, 15, 30, 45	deg	A	Plate tilts about H and about V
Capture control				
WARMUP_MIN_MINUTES	45	min	4	Minimum powered time before any capture
WARMUP_DRIFT_WINDOW_MIN	10	min	4	Window over which drift is judged

WARMUP_DRIFT_FRACTION_OF_SIGMA	0.1	—	4	Allowed drift of the mean plane Z, as a fraction of σ_t at that Z
DRIFT_SENTINEL_INTERVAL_MIN	60	min	all	Interval between sentinel captures
ROBOT_SETTLE_TIME_S	2.0	s	all	Wait after motion before capture; Step 4.4 verifies it
FRAMES_PER_NOISE_STATION	100	frames	A	Frames per Z, field, and tilt pose
FRAMES_PER_EDGE_POSE	30	frames	B-HV	Frames per edge pose
FRAMES_PER_ZSTEP_POSE	10	frames	B-Z	Frames per Z-step pose
FRAMES_PER_AREA_POSE	10	frames	C	Frames per area pose
FRAMES_PER_DETECTION_TRIAL	1	frames	D	One frame per trial, giving single-frame detectability
Phase randomization				
PHASE_JITTER_SPAN_PX	8	px at the station Z	B, C, D	Square span of random lateral offsets, larger than the projector dot pitch and the noise correlation length
PHASE_JITTER_POSES_EDGE	25	poses	B-HV	Offsets per edge station
PHASE_JITTER_POSES_AREA	30	poses	C	Offsets per area station
Z-step test				
Z_STEP_LADDER_MM	0.1, 0.2, 0.5, 1, 2, 4	mm	B-Z	Commanded step sizes
Z_STEP_REPEATS	10	ABAB cycles	B-Z	Alternations for each step size
Z_STAIRCASE_SUBDIVISION	10	per quantum	B-Z	Fine-sweep points per expected quantum
ROBOT_REPEATABILITY_MM	0.05	mm	B-Z	Position repeatability of the robot (ISO 9283, the Section 3.1 requirement). The read-back pose is the step truth of B-Z, so a rung smaller than this is captured but flagged: its truth is no better than the robot itself
Z_STEP_APPROACH_OVERSHOOT_MM	2	mm	B-Z	Every Z visit is approached from below by this overshoot and a final move

				up onto the pose, so that backlash and compliance do not enter the A → B difference
Detection (D, also used in B-Z)				
DETECTION_FALSE_ALARM_TARGET	0.01	per window	B-Z, D	Sets threshold τ from the blank-site distribution
DETECTION_MIN_CONNECTED_PX	2	px	D	Minimum connected region counted as a detection
DETECTION_WINDOW_MARGIN_PX	3	px	D	Search window radius equals D/2 in px plus this margin
DETECTION_LEVELS	9	diameters	D	Diameter levels per curve
DETECTION_LEVEL_LOW_FACTOR / HIGH_FACTOR	0.3 / 2.5	× pilot D ₅₀	D	Range spanned by the levels, wide enough to include the 0% and 10% points
DETECTION_TRIALS_PER_LEVEL	60	trials	D	Independent trials per level
DETECTION_ZERO_TRIALS	300	trials	D	Trials at the levels next to the 0% point
DETECTION_ZERO_PROBABILITY_BOUND	0.01	—	D	Upper bound that defines practically zero detection
CONFIDENCE_LEVEL	0.95	—	B, C, D, E	For all confidence intervals and bounds
BOOTSTRAP_RESAMPLES	2000	—	C, D, E	Resamples for confidence intervals
Boundary bias (E)				
BOUNDARY_BAND_HALF_WIDTH_PX	8	px	E	Analysis band on each side of a true edge
BOUNDARY_BIN_WIDTH_PX	0.25	px	B, E	Signed-distance bin width, giving 4× oversampling
SURFACE_ASSIGNMENT_SIGMA_MULTIPLE	3	× $\sigma_{\text{tot}}(Z)$	E	A read within this distance of a reference plane is assigned to that plane
Targets				
NOISE_PLATE_SIZE_MM	400 × 400	mm	A, B-Z	Uniform matte plate
PLATE_FLATNESS_MM	0.05	mm	all	Must be at most 0.25 × the smallest expected σ_{tot}
GAP_SMALL_MM / GAP_LARGE_MM	15 / 60	mm		

			B, C, D, E	Distance from front plate to back plate
EDGE_SLANT_DEG	5	deg	B-HV	Edge rotation from the image axes
EDGE_SQUARE_SIZE_MM	160	mm	B-HV	Side of the raised square and of the square window
CHAMFER_MARGIN_DEG	10	deg	3	Margin added to the worst- case ray angle
EDGE_LAND_MAX_MM	0.2	mm	3	Maximum residual flat land at a knife edge
DIAMETER_LADDER_RATIO	$\sqrt{2}$	—	C, D	Ratio between successive diameters
DIAMETER_MIN_FOOTPRINT_FRACTION	0.3	$\times$ $p(Z_{\text{MIN}})$	C, D	Smallest diameter, as a fraction of the pixel footprint at Z_{MIN}
DIAMETER_MAX_FOOTPRINT_MULTIPLE	30	$\times$ $p(Z_{\text{MAX}})$	C	Largest diameter, as a multiple of the pixel footprint at Z_{MAX}
FEATURE_ISOLATION_PX	30	px at Z_{MIN}	C, D	Minimum edge-to-edge spacing between features
POST_DIAMETER_FRACTION_OF_D0	0.5	$\times$ pilot D_0	3	Disk support posts must be thinner than this
Registration				
REGISTRATION_POSES	30	poses	4	Hand-eye poses spanning the volume
REGISTRATION_TILT_RANGE_DEG	± 20	deg	4	Tilt range about H and V
REGISTRATION_FRAMES	10	frames	4	Frames per registration pose
REGISTRATION_RESIDUAL_ACCEPT_MM	0.15	mm RMS	4	Acceptance limit for the registration residual

These relations are used throughout. $p(Z)$ is the pixel footprint, θ the subtended angle, d the disparity, and δZ_q the depth quantum implied by a disparity quantum q :

$$p(Z) = \frac{Z}{f_x}, \quad \theta = \frac{D}{Z}, \quad D_{\text{px}} = \frac{D}{f_x} Z, \\ \quad d = \frac{k}{Z}, \quad k = f_x B, \quad \delta Z_q = \frac{q}{Z^2} k$$